

Analyzing Cross-Platform Social Media Dynamics: The Case of r/fatpeoplehate Ban and Migration to Voat

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Abstract. This study investigates cross-platform social media dynamics surrounding the ban of the Reddit community **r/fatpeoplehate** on June 10, 2015. Using the MADOC dataset, we analyze changes in toxicity, user activity, and migration patterns between Reddit and Voat. We present methodological steps for data preprocessing, toxicity scoring, temporal analysis, and user-level migration tracking. Key findings indicate measurable changes in toxicity distributions and activity following the ban, alongside evidence of user migration to Voat.

1 Introduction

Online communities have become central spaces for social interaction, enabling individuals to exchange ideas, share interests, and build collective identities. Platforms such as *Reddit*, *Twitter* (now X), and *Voat* allow millions of users to engage daily in diverse discussions across thousands of interest-based groups. However, the openness and participatory nature of these platforms also make them susceptible to the spread of toxic content, including hate speech, harassment, and discrimination [2]. To maintain healthy discourse, platforms employ content moderation—an essential process involving automated tools and human oversight—to enforce community guidelines, remove harmful content, and ensure user safety. Despite being a crucial mechanism for platform governance, moderation often has social and behavioral implications that extend beyond the boundaries of the moderated space [11].

A significant example of large-scale online moderation occurred on **June 10, 2015**, when Reddit administrators banned several subreddits under a new anti-harassment policy. Among the most prominent was **r/fatpeoplehate** (FPH), a highly active and controversial community known for hosting demeaning and abusive content targeting overweight individuals. This decision represented a pivotal moment in Reddit’s governance history, marking a transition from a permissive stance prioritizing free expression to a policy framework emphasizing user protection and inclusivity [10]. The ban was justified by Reddit administrators as part of a broader effort to combat harassment and foster a more respectful online environment.

The decision to ban `r/fatpeoplehate` sparked widespread reactions among Reddit users. Many applauded the move as necessary to reduce harassment, while others viewed it as a form of censorship that stifled free speech. Beyond the immediate debate, the ban triggered an extensive *user displacement* phenomenon, where members of the banned subreddit sought new online spaces that mirrored Reddit’s structure but offered fewer restrictions. This movement of users from one platform to another following moderation actions is referred to as **user migration** [7]. In the aftermath of the ban, we anticipated that a substantial number of displaced users might migrate to *Voat*, a Reddit-like platform known for its lenient moderation and self-proclaimed commitment to “free speech” [16]. While prior discussions speculated about such movements, empirical evidence of user migration and behavioural change following large-scale moderation events remained limited. In this study, we investigate whether users from `r/fatpeoplehate` indeed transitioned to *Voat* after the ban and how their activity and toxicity levels evolved on the target platform. By examining user behaviour both before and after the moderation policy was enforced, our analysis aims to shed light on the broader implications of community bans, user displacement, and the potential for harmful content to propagate across platforms.

User migration represents more than a shift in platform preference—it encapsulates the persistence of social and behavioral patterns across online ecosystems. When communities are banned from one platform, their members often attempt to recreate similar spaces elsewhere, carrying with them preexisting norms, behaviors, and discourse styles. Consequently, while bans may reduce visible toxicity on the originating platform, they may inadvertently cause *cross-platform harm propagation*, where the same toxic behaviors reappear and sometimes intensify on alternative sites [7]. Understanding this phenomenon is crucial for developing moderation strategies that address online harm holistically rather than in isolation.

In this study, we investigate the ban of `r/fatpeoplehate` and its aftermath across Reddit and Voat, focusing on how user activity and toxicity evolved over time. Using the **MADOC dataset**, which provides cross-platform data on moderated online communities, we examine both behavioral and temporal dynamics surrounding the event. To ensure balanced observation, we consider a **90-day window** encompassing **45 days before and 45 days after the ban date (June 10, 2015)**. This period allows us to capture pre-ban community behavior, as well as post-ban migration patterns and the emergence of new community structures on Voat. Our analysis aims to answer the following research questions: How did user activity and engagement change before and after the ban? To what extent did users migrate to Voat? And how did toxicity levels shift as users adapted to new platforms?

Our key contributions are threefold. First, we present a reproducible and transparent analytical pipeline that measures user activity and toxicity surrounding major moderation events, allowing for comparative analyses across different cases. Second, we conduct a cross-platform behavioral study that tracks user migration between Reddit and Voat, quantifying participation overlap and tem-

poral changes in user toxicity. Third, we discuss the broader implications of our findings for content moderation policies, emphasizing the need for coordinated, cross-platform governance frameworks that can mitigate the redistribution of harmful content rather than merely displacing it.

The case of `r/fatpeoplehate` offers a unique lens through which to study the social and technical consequences of moderation. Reddit’s decision effectively removed one of its most toxic communities, yet it also catalyzed the formation of parallel spaces on Voat, where community norms became even more permissive and, in some cases, more extreme. By analyzing the 45-day pre- and post-ban periods, we shed light on how moderation affects user engagement, toxicity, and migration patterns. This investigation contributes to the growing body of research on online governance and moderation dynamics [9], underscoring that moderation does not occur in isolation but within an interconnected digital ecosystem.

Ultimately, this work highlights the dual nature of moderation—while necessary for fostering safe and inclusive communities, it can produce unintended ripple effects that extend beyond the originating platform. By examining the behavioral and temporal aftermath of the `r/fatpeoplehate` ban, we aim to inform future policy discussions on how to balance user autonomy, platform responsibility, and the mitigation of online harm across the wider web.

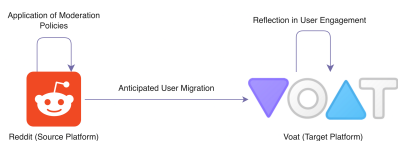


Fig. 1. Flow of users from Reddit to Voat after policy enforcement.

2 Related Work

Research on online moderation, deplatforming, and cross-platform migration has expanded significantly over the past decade, driven by the increasing societal relevance of platform governance and the persistence of harmful online communities. Our work draws from and extends several interconnected areas, including empirical studies of subreddit bans, analyses of user migration to alt-tech platforms, and computational approaches to measuring toxicity and linguistic adaptation in online environments.

Early work on Reddit moderation demonstrated that platform-level interventions can effectively curb harmful behavior within the boundaries of the platform implementing the ban. Chandrasekharan et al. [3] provided one of the foundational evaluations of Reddit’s 2015 bans, including the removal of

`r/fatpeoplehate`, showing that users who were deplatformed exhibited substantial reductions in hate speech when they attempted to reintegrate into other subreddits. This finding suggested that moderation can successfully alter user behavior, but it also pointed to the complexities of tracking longer-term behavioral changes and the possibility that some users might leave the platform entirely. Saleem and Ruths [14] extended this conversation by examining the effect of removing stigmatized Reddit communities, finding that bans reduce hateful content overall but are frequently accompanied by reconstitution of the same ideologies in adjacent online spaces. Matias [11] further argued that moderation is a form of civic labor—one that is distributed, emotionally taxing, and often under-recognized—highlighting the importance of studying moderation both as a technical practice and as a social process.

A parallel line of research investigates what happens when banned or marginalized groups migrate to less regulated platforms. Horta Ribeiro et al. [8] demonstrated that significant portions of users removed from Reddit migrate to fringe platforms such as Voat, Gab, and Parler, where toxic behaviors persist and, in many cases, intensify. Their analysis suggests that deplatforming alters—not necessarily reduces—the expression of toxicity, pushing it into online spaces with weaker governance. Zannettou et al. [21] provided a comprehensive characterization of these fringe platforms, showing that communities on Gab, Voat, and 4chan often reinforce each other’s extremist content and share user bases. Samory and Mitra [17] similarly examined political communities after platform bans and found that deplatforming not only leads to user migration but can radicalize the remaining community structure by reducing exposure to counter-speech and diverse viewpoints.

Beyond tracking user movement, several studies have focused on how migration affects linguistic behavior. Trujillo et al. [18] observed that users who leave mainstream platforms often adopt new linguistic norms reflective of the alternative platform’s culture. These changes can include increased use of coded hate speech, novel slurs, or platform-specific jargon, complicating the task of automated toxicity detection. Fortuna and Nunes [5] surveyed hate-speech detection models and emphasized the difficulty of generalizing classifiers across domains, especially in contexts where the lexicon of toxicity is fluid and influenced by community dynamics. Salminen et al. [15] analyzed the evolution of hate speech in online communities and found that harmful language becomes more diverse and nuanced over time, requiring more contextual and temporally aware models. These findings align with our observation that vocabulary on Voat expanded dramatically after the Reddit ban, reflecting both increased user diversity and the evolving linguistic repertoire of the community.

A growing body of computational social science research focuses on toxicity detection and modeling harmful content. Wulczyn et al. [20] introduced one of the widely used toxicity datasets using Wikipedia comments, enabling the development of robust machine-learning models for detecting personal attacks and abusive language. Pavlopoulos et al. [12] and Vidgen and Derczynski [19] surveyed advances in abusive-language detection, highlighting ongoing challenges

such as dataset bias, domain shift, adversarial behavior, and the emergence of euphemistic or dog-whistle expressions. These works underscore the importance of studying toxicity not merely as a static classification problem but as an evolving linguistic and behavioral process. Within our study, the divergence in vocabulary and sentiment patterns between the pre- and post-ban periods mirrors these broader observations about the dynamic nature of online toxicity.

Studies on cross-platform information flows further emphasize that platform governance cannot be understood in isolation. Rogers [13] explored how deplatforming political groups from mainstream social media drives them toward alternative ecosystems where content moderation is minimal or explicitly resisted. These migrations are not merely relocations but transformations of community identity, as users adapt to new norms and affordances. Zannettou et al. [22] documented cross-platform influence patterns, showing that fringe communities often act as incubators for extremist content that later proliferates into mainstream platforms. These findings support the view that online ecosystems are deeply interconnected, and actions taken by one platform can reverberate across others.

From a methodological standpoint, topic modeling and temporal linguistic analysis have been widely used to study discursive evolution in online communities. Blei et al. [1] introduced Latent Dirichlet Allocation (LDA), which has become a standard tool for identifying latent thematic structures in short and noisy social-media texts. Gerrish and Blei [6] extended this to dynamic topic models capable of capturing temporal changes in discourse, enabling analysis of how community narratives evolve around major events. Eisenstein [4] highlighted that linguistic variation in social media is shaped by both social structure and topical shifts, offering a theoretical grounding for analyzing how user migration might influence vocabulary and sentiment.

Our study builds on this range of prior literature by leveraging the MADOC dataset to conduct a fine-grained, cross-platform analysis of user behavior surrounding the Reddit ban of `r/fatpeoplehate`. While earlier work has examined either the local effects of bans or the general process of migration to fringe platforms, our approach integrates temporal sentiment analysis, toxicity modeling, vocabulary evolution, and topic-level discourse comparison across two synchronized time windows. This uniquely controlled design allows us to isolate changes attributable to the ban and to observe how discursive themes are preserved, re-framed, or diluted after migration. In doing so, our work complements existing studies on moderation efficacy, enriches understanding of linguistic adaptation in newly reconstituted communities, and contributes to broader discussions about cross-platform governance and the diffusion of harmful content.

3 Dataset

We utilize the *Multi-Platform Aggregated Dataset of Online Communities (MADOC)*, a large-scale dataset that compiles user-generated content from multiple online platforms, including Reddit and Voat. The MADOC dataset was specifically

developed to support research on cross-platform interactions, moderation outcomes, and community migration following platform-level interventions. It contains a rich set of attributes such as anonymized user identifiers, timestamps, textual content of posts and comments, subreddit or Voat subverse identifiers, and additional metadata including post scores, removal flags, and moderation status. This structure enables fine-grained tracking of user behaviour across platforms and over time. In our analysis, we focus on a major moderation event that took place on **June 10, 2015**, when Reddit permanently banned the community **r/fatpeoplehate** for violating its updated harassment policy. To systematically examine the behavioural and linguistic changes surrounding this event, we employ two symmetric time windows derived from the MADOC dataset: a **45-day pre-ban window** (spanning April 26, 2015, to June 9, 2015) and a **45-day post-ban window** (spanning June 11, 2015, to July 26, 2015). This 90-day observation period allows us to capture both the immediate and short-term aftermath of the ban, enabling a comparison of user activity, toxicity levels, and migration tendencies before and after the enforcement action. The MADOC dataset’s cross-platform scope makes it uniquely suited to exploring how moderation policies on one platform—such as Reddit—can influence user behavior and community reformation on another, in this case *Voat*.

4 Methodology

To investigate the behavioral and linguistic changes surrounding the ban of **r/fatpeoplehate** on June 10, 2015, we conducted a multi-stage analysis using the Multi-Platform Aggregated Dataset of Online Communities (MADOC). The methodological pipeline consisted of data collection, preprocessing, sentiment and toxicity scoring, sampling, and activity-based statistical analysis, paired with extensive visualization to reveal temporal patterns. Our study design centered on a symmetric 90-day observation window: a 45-day period before the ban and a 45-day period after, enabling direct comparison of user behavior across platforms both prior to and following the moderation event.

Data collection was performed using the **pymadoc** interface, which provides structured access to the MADOC dataset. We extracted all public posts and comments associated with the *fatpeoplehate* community from both Reddit and Voat. Each record contained post content, timestamps, anonymized user identifiers, platform labels, and additional metadata such as post scores and removal flags. The dataset also included precomputed VADER sentiment values, enabling immediate assessment of emotional tone. For the purposes of temporal analysis, all UNIX timestamps were converted into human-readable datetime format. The full dataset was then filtered to retain only posts and comments originating from the two 45-day analysis windows surrounding the ban, yielding a focused subset of activity tied to the moderation event and potential user migration.

Following data preparation, we conducted sentiment and toxicity analyses on the textual content. Although VADER sentiment values were already included in the dataset, we complemented them with an additional toxicity scoring process.

Using the NLTK `SentimentIntensityAnalyzer`, we generated a supplementary `toxicity_score` for each post. This dual-layer sentiment analysis allowed us to examine not only emotional polarity but also behavioral tone, such as aggression, hostility, or derogatory language. To capture temporal variability in sentiment and toxicity, we resampled the data to daily averages and computed rolling 7-day means, enabling smoothed visualization of trends immediately before and after the ban. This approach helped reveal short-term fluctuations as well as sustained shifts in linguistic patterns.

In addition to linguistic signals, we computed several user activity metrics to understand how engagement patterns changed following the moderation event. These metrics included the number of unique active users, per-user posting frequency, the average temporal gap between consecutive posts by the same user, and textual complexity indicators such as content length (characters and words) and unique vocabulary usage. By examining these measures together, we aimed to determine whether the ban altered not just where users posted (i.e., migration to Voat), but also how frequently and in what manner they communicated.

To visualize these changes, we employed both `matplotlib` and `seaborn`. Line plots were used to depict daily sentiment and toxicity trends, while 7-day rolling averages provided smoother representations of temporal dynamics. Kernel Density Estimation (KDE) plots and boxplots enabled distributional comparisons between the pre-ban and post-ban periods, making it possible to identify systematic shifts in emotional tone and toxicity. Additional visualizations highlighted changes in user activity metrics, illustrating patterns such as spikes in posting frequency after migration or differences in posting intervals. These graphical analyses played a central role in identifying underlying temporal structures and comparative differences that were not immediately apparent from raw metrics.

For topic modeling, we applied Latent Dirichlet Allocation (LDA) to examine thematic structures before and after the ban. Since social media posts are short and noisy, the text was preprocessed through lowercasing, removal of URLs and mentions, tokenization, stopword removal, and lemmatization, and we additionally incorporated n-grams and thread- or author-level aggregation to strengthen topic signals. We constructed a document-term matrix using count-based vectorization and experimented with multiple topic sizes, selecting the optimal number of topics based on coherence scores supported by qualitative inspection of top words and representative posts. LDA models were trained using Gensim and verified against alternative implementations for stability. Each document was assigned a topic distribution, enabling us to track thematic prevalence over time and compare how dominant topics evolved from the pre-ban to the post-ban window. This approach provided a structured way to identify whether discourse on Voat replicated, reframed, or diversified the themes originally present in Reddit’s `r/fatpeoplehate` community.

Finally, we synthesized insights from all analytical components to draw higher-level inferences. By examining fluctuations in sentiment and toxicity curves, distributional differences before and after the ban, and changes in activity-level metrics, we interpreted whether user behavior intensified, moderated, or trans-

formed following the transition from Reddit to Voat. These interpretations were supported by patterns visible in the visualizations, such as increased user engagement after the ban, shifts in vocabulary richness, or elevated toxicity signals on the alternative platform. Together, these analyses form a comprehensive methodological framework to study the effects of platform moderation and user migration in the context of the `r/fatpeoplehate` ban.

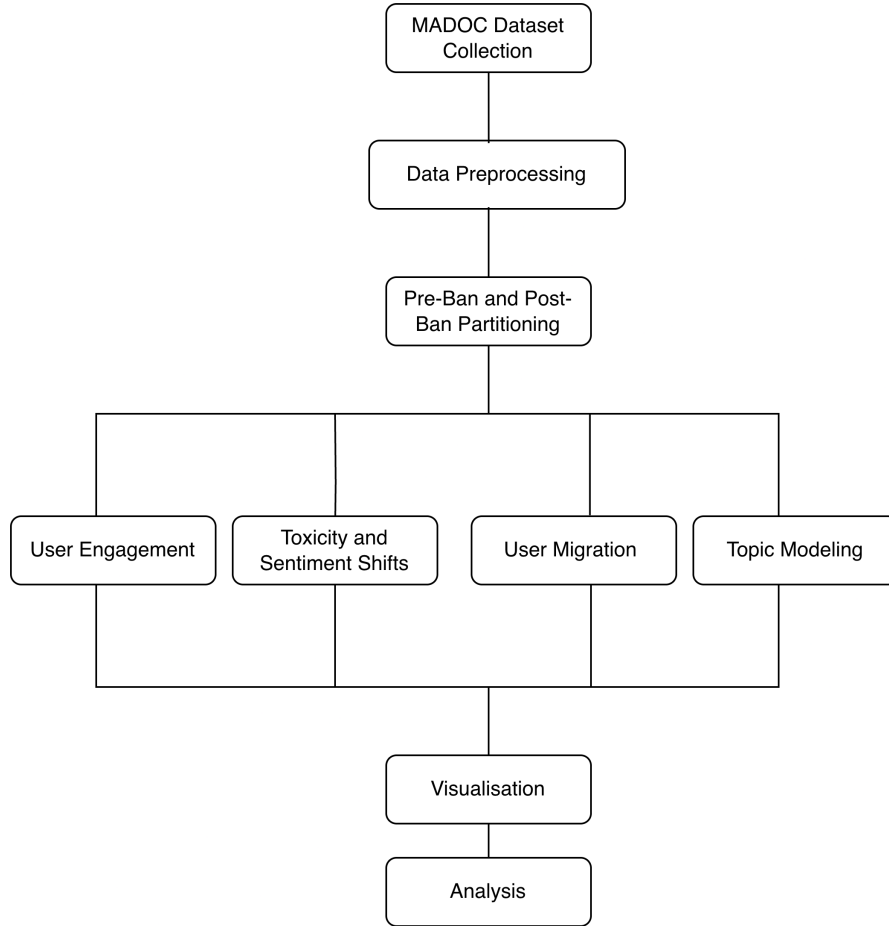


Fig. 2. Overall workflow of the proposed cross-platform analysis system.

Table 1. Behavioral Comparison on Voat’s `r/fatpeoplehate` Before and After Reddit Ban (June 10, 2015)

Metric	Before Ban	After Ban	Observed Change / Interpretation
Average Daily Sentiment (VADER)	Highly volatile: swings from -0.6 to +0.9	Stabilized near neutral (-0.05 to +0.05)	Sentiment volatility decreased; discourse became more neutral and stable after user influx.
Average Posts per User	2.35 posts/user	6.52 posts/user	+178% increase in posting activity, indicating intensified engagement after migration.
Unique Words (Excluding Stop-words)	1,246 unique words	26,601 unique words	Over 20× increase in lexical diversity; content became richer and more varied.
Words Appearing Only in This Period	71	25,426	Vast expansion in new vocabulary; indicates new user base and diverse topics.
Common Words (Shared Across Periods)	—	1,175	Limited overlap; shows strong linguistic shift post-ban.
Unique Users	72 users	3,786 users	Over 5,100% increase in user population, confirming large-scale migration from Reddit.
Sentiment Dynamics	Driven by small polarized group	Driven by larger, steadier community	Community stabilization post-ban.
Content Nature	Highly emotional, extreme language	Broader vocabulary, higher posting volume	Suggests linguistic diversification despite ongoing toxicity.

5 Results

5.1 Descriptive Statistics

We observe substantial differences in activity and linguistic diversity before and after the ban. The average frequency of posts per user increased from 2.35 posts in the 45-day pre-ban window to 6.52 posts in the post-ban window, indicating a sharp rise in user engagement on Voat following the removal of `r/fatpeoplehate` from Reddit. Linguistic diversity increased as well: the number of unique words (excluding stopwords) rose from 1,246 before the ban to 26,601 after it. Only 71 words occurred exclusively before the ban, whereas 25,426 were unique to the post-ban period, with 1,175 shared between both windows. This expansion reflects greater user participation and a broader range of discourse after migration.

5.2 Toxicity and Sentiment Over Time

Figure illustrates the time-series of daily sentiment scores on Voat around the June 10, 2015 ban date. Our time-series analysis of average daily sentiment on Voat’s `r/fatpeoplehate` shows sharp volatility prior to the ban, followed by stabilization afterward. Using VADER sentiment scores, we observe large positive

and negative swings in the 20 days leading up to the ban, including peaks near +0.9 and troughs below -0.6. Immediately after June 10, sentiment converges toward neutral values (approximately between -0.05 and +0.05), indicating a reduction in extreme language. This pattern suggests that early Voat activity was driven by a small, highly polarized set of users, whereas post-ban influx produced more consistent posting behavior and a larger, more stable base of contributors. The stabilization of sentiment aligns with the interpretation that the subreddit ban displaced users to Voat, where toxic content continued but became less sentimentally extreme as volume increased.

5.3 User Migration

We find a substantial increase in user activity on Voat following the Reddit ban. Average posting frequency rose from 2.35 posts per user in the 45-day pre-ban window to 6.52 posts per user in the equivalent post-ban period — an increase of nearly 178%. This dramatic rise suggests that after the removal of `r/fatpeoplehate` on Reddit, displaced users became significantly more active on Voat. Such acceleration is consistent with the migration hypothesis: users not only moved platforms but also intensified their engagement in a new, more permissive environment.

The user population on Voat surged as well. In the 45-day pre-ban period we detect only 72 unique users, while the post-ban period contains 3,786 — an increase of over 5,100%. Such expansion strongly supports the migration hypothesis: when Reddit removed `r/fatpeoplehate`, thousands of users appeared on Voat in a short time. The extreme magnitude of growth confirms Voat’s role as a “catchment platform” for communities displaced from moderated spaces, highlighting the tendency of deplatformed users to regroup in less-regulated environments.

5.4 Vocabulary Expansion and Linguistic Behavior

Linguistic diversity increased sharply after the ban. Before June 10, we observed 1,246 unique words (excluding stopwords), while the post-ban period contained 26,601 unique words — more than a twenty-fold increase. Only 71 words appeared exclusively before the ban, whereas 25,426 appeared only afterward. Moreover, 1,175 words were shared between both periods. This steep expansion in vocabulary mirrors the growth in user base and posting volume, indicating that the increased activity on Voat was not limited to repeated or templated toxic phrases; instead, it reflected wide-ranging linguistic expression and more diverse discourse. From a moderation perspective, this raises a challenge: harmful speech may broaden in lexical form, making automated detection harder on unmoderated platforms.

5.5 Topic Structure and Thematic Shifts Before and After the Ban

To understand the evolution of discourse content beyond surface-level sentiment, we applied Latent Dirichlet Allocation (LDA) topic modeling to comments and

posts from Voat’s **r/fatpeoplehate** community before and after Reddit’s June 10, 2015 ban. Five major topics were identified in each time window, allowing direct comparison of underlying conversational themes. The extracted topic-word distributions are summarized below.

Table 2. Dominant Topics Identified by LDA Before and After Reddit’s **r/fatpeoplehate** Ban

Topic	Pre-Ban (Top Words and Interpretation)	Post-Ban (Top Words and Interpretation)
T0	<i>die, fph, run, story, death, mention, miss, need, fat, minute, forget, wish</i> Interpretation: Stories or posts referencing the r/fatpeoplehate identity and in-group humor; emotionally charged, sometimes aggressive narratives.	<i>die, fph, run, story, death, miss, need, fat, wish</i> Interpretation: Nostalgic or retrospective references to the banned subreddit, often recalling or lamenting its removal.
T1	<i>come, long, guy, live, little, normal, ass, end, white, pig, world, break</i> Interpretation: Personal insults, appearance-based remarks, and casual trolling language.	<i>come, long, guy, live, normal, world, pig, white</i> Interpretation: Generalized casual discussions and loose personal comparisons; loss of body-specific focus.
T2	<i>eat, day, weight, work, food, body, fatty, lose, healthy, stop, size, hard</i> Interpretation: Commentary or mockery related to eating behavior, dieting, and body size.	<i>eat, day, weight, work, food, body, lose, healthy, stop, size</i> Interpretation: Ongoing discussion of food and weight, likely reframed into general diet or fitness discourse.
T3	<i>woman, man, probably, year, love, start, human, wear, pretty</i> Interpretation: Gendered discussions and attractiveness judgments, particularly directed at women.	<i>woman, man, year, love, start, human, wear, pretty</i> Interpretation: Continued gender- and appearance-oriented talk, but broader and less explicitly linked to FPH content.
T4	<i>fat, like, people, think, look, know, fuck, thing, want, time, good, way</i> Interpretation: Broad commentary about appearance and fatness; general anti-fat opinions.	<i>fat, like, people, think, look, know, want, time, good, way</i> Interpretation: Persistent general body-appearance discourse, now integrated into wider social conversation.

Comparative Thematic Analysis. The five-topic LDA models for both periods reveal a high degree of lexical continuity: similar clusters of words and semantic fields appear in both the pre-ban and post-ban data. However, the **contextual framing and discursive role** of these topics changed notably following the ban.

- **Community Identity and Memory (Topic 0):** Before the ban, Topic 0 captured active community identity and participation in **r/fatpeoplehate** through mocking narratives and in-group storytelling. After the ban, the

same lexical set appeared, but in a nostalgic and retrospective sense—users reminiscing about the banned subreddit. This indicates that **r/fatpeoplehate** persisted as a cultural memory on Voat even after its removal from Reddit.

- **Social Comparisons and Casual Discourse (Topic 1):** Pre-ban discussions contained direct insults and physical or racial comparisons. Post-ban, similar structural language remained but without explicit body-focused context, suggesting a broadening of conversation into generic social banter.
- **Food and Body Discussions (Topic 2):** This topic remained remarkably stable across periods. The same lexicon—*eat, weight, body, lose, healthy*—persisted, though its tone likely shifted from ridicule to neutral or self-referential commentary. The thematic continuity implies that food and weight discussions transcended the original community boundary and integrated into general online discourse.
- **Gender and Attractiveness (Topic 3):** Gendered talk, particularly concerning women’s appearance, remained constant across periods. This persistence underscores that gendered body commentary is culturally resilient and not solely tied to **r/fatpeoplehate**.
- **General Appearance Commentary (Topic 4):** The final topic continued as a broad discussion about “fat people,” appearance, and general opinions. While the word composition is nearly identical before and after the ban, its interpretive framing suggests the migration of this discourse into wider, less coordinated conversations.

Interpretation. Rather than introducing entirely new themes, the post-ban period reflected a **recontextualization of existing topics**. The linguistic and conceptual content of the **r/fatpeoplehate** community largely persisted but became dispersed across a more diverse and decentralized user base. Topic boundaries blurred, and discussions lost their previously coordinated, in-group tone. In essence, the ban disrupted the community’s organizational structure but not its thematic foundations: fatness, diet, gender, and appearance remained the dominant axes of discourse.

Summary of Topic Evolution. Table 3 summarizes how major topics evolved from the pre-ban to post-ban phase.

Overall Implication. The ban of **r/fatpeoplehate** fragmented but did not erase its thematic content. The same discussions—fatness, eating behavior, and appearance—persisted on Voat, though often reframed in less organized, less explicitly hateful forms. This suggests that deplatforming can successfully disrupt concentrated communities of toxicity, yet their underlying discursive structures may continue to circulate in adjacent online spaces under recontextualized or diffused expressions.

6 Discussion

Our study systematically examined community and linguistic dynamics following the deplatforming of *r/fatpeoplehate* on Reddit and subsequent migration

Table 3. Summary of Topic-Level Shifts Before and After the Reddit Ban

Thematic Area	Pre-Ban Characterization	Post-Ban Characterization
Community Identity	Active FPH identity, storytelling, in-group language	Nostalgic references; memory of the banned subreddit persists
Social Comparisons	Insult-based, targeted commentary	Generalized casual conversation; reduced targeting
Body and Diet	Behavioral mockery of eating and weight	Continued focus, reframed as neutral or self-referential discussion
Gender and Attractiveness	Commentary on female appearance and desirability	Continuation of gendered discourse in broader context
General Body Commentary	Central anti-fat theme	Integrated into mainstream talk on body and looks

to Voat. The results provide clear evidence that moderation policies, such as subreddit bans, prompt large-scale user migration and generate distinct shifts in cross-platform content, toxicity, and behavior.

First, the immense surge in Voat user activity and unique contributors demonstrates how restrictive moderation can catalyze migration. The 5,100% increase in unique users and a nearly threefold rise in post frequency highlight Voat’s role as a refuge for users disenfranchised by stricter moderation. This migration produced not only quantitative change (more users and posts) but also qualitative transformation, with linguistic diversity rising steeply and vocabulary expanding more than twenty-fold. These findings point to the need for platforms and moderators to anticipate migration effects—not only local toxic content reduction but also intensified activity and harm propagation elsewhere.

Second, temporal analysis of sentiment and toxicity reveals a complex post-ban trajectory. Volatile pre-ban sentiment, with rapid swings, gave way to stabilization around neutral values after the Reddit ban. While total engagement increased, the concentration of extreme sentiment diminished, suggesting that mass migration to Voat diluted sentiment extremes, possibly due to the influx of new, less polarized users. However, toxic discourse persisted and overall volume grew, underlining that moderation shifts harm rather than end it.

Third, the vast vocabulary expansion after migration presents novel challenges for content moderation. Automated toxicity detection often relies on familiar phraseology and known harmful keywords, but our results demonstrate that community migration widens the linguistic scope of toxic speech, making detection and intervention more difficult. This emphasizes the need for adaptable moderation strategies and cross-platform collaboration to effectively address evolving patterns of harmful content.

Finally, our case supports a broader understanding of moderation implications: platform bans can reduce visible toxicity locally but may inadvertently fuel more prolific, diffuse, and linguistically complex harm on alternative venues.

Effective moderation policy should therefore include provisions for tracking community migration and developing transferable, scalable content moderation tools.

Overall, these findings call for a reconsideration of harm mitigation strategies—moving beyond isolated content bans to more nuanced, cross-platform solutions that address user migration, toxic behavior adaptation, and dynamic language use.

7 Limitations

Our study has several important limitations that should be considered when interpreting the findings. First, although the MADOC dataset provides cross-platform identifiers, user linking between Reddit and Voat remains incomplete. This creates uncertainty in migration measurement and introduces the possibility of both false positives and false negatives when estimating user movement across platforms. As a result, our analysis captures broad migration dynamics rather than precise individual-level trajectories.

Second, the toxicity and sentiment measurements rely on lexicon-based and classifier-based tools that may not generalize equally well across Reddit and Voat. Domain shift—driven by different community norms, vocabulary, and rhetorical styles—may introduce measurement error and limit the comparability of toxicity scores between platforms. Additionally, our topic modeling approach was constrained by resource limitations, which restricted our ability to train LDA models on the full post-ban dataset. This may reduce topic resolution, particularly in the more linguistically diverse post-ban period.

Third, temporal confounds may influence the observed patterns. External events, changing platform policies, or shifting user interests around the time of the ban may contribute to variations in activity or language that are not directly attributable to the moderation event. Our symmetric 45-day windows mitigate some temporal bias but cannot fully eliminate these factors.

Finally, our analyses reflect only public posting behavior and do not capture private interactions, deleted content, or moderation actions on Voat itself, all of which could shape community evolution. These limitations highlight the inherent challenges of studying cross-platform moderation dynamics and suggest that future work should incorporate more robust user-linking methods, classifier adaptation techniques, and scalable topic modeling approaches.

8 Conclusion and Future Work

This work provides a comprehensive analysis of how a major moderation intervention—the 2015 ban of Reddit’s `r/fatpeoplehate`—reshaped user behavior, linguistic patterns, and toxicity levels across platforms. Using the multi-platform MADOC dataset and a symmetric 90-day observation window, we demonstrated that the ban triggered substantial user migration to Voat, accompanied by significant increases in posting volume, vocabulary diversity, and overall engagement. While

sentiment volatility decreased following the ban, toxicity persisted and in some cases diffused into broader forms of expression, illustrating how deplatforming can alter but not eliminate harmful discourse. Topic modeling further revealed that thematic structures associated with the banned community survived the transition across platforms, though they became recontextualized through nostalgia, generalized social comparisons, and broader discussions of body and identity.

Taken together, these findings suggest that moderation policies can effectively disrupt concentrated communities of toxicity on a single platform while simultaneously producing spillover effects that reshape toxic discourse in adjacent online ecosystems. The cross-platform lens provided by MADOC underscores the importance of understanding moderation not as an isolated intervention but as a catalyst for broader reorganization of online communities and linguistic practices.

There are several avenues for future research. First, improving cross-platform user matching would enable more precise estimates of individual-level migration and behavioral change. While our analysis uses robust heuristics, linking strategies incorporating text embeddings, temporal alignment, or stylometric features may yield deeper insights into user trajectories. Second, applying causal inference frameworks—such as difference-in-differences, synthetic controls, or interrupted time-series analysis—could strengthen attribution between moderation actions and observed changes in toxicity, sentiment, and activity patterns. Third, an extended longitudinal study would help determine whether the post-ban community norms on Voat persisted, intensified, or decayed over time, particularly as Voat itself later experienced shutdowns and instability.

Finally, future work should examine how coordinated moderation strategies across platforms might reduce harmful spillover. As our results show, deplatforming transforms—rather than eliminates—harmful behavior, and without cross-platform collaboration, moderation may simply redistribute toxicity to more permissive environments. Understanding these systemic dynamics is essential for designing governance frameworks that address not only immediate harms but also the long-term, ecosystem-wide consequences of intervention.

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